

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS Computer Science candidate (NC State, May 2026) with hands-on experience in systems programming, vector index infrastructure, and fault-tolerant distributed systems. Engineered a virtual memory system in C for Xinu OS -- 8-frame page store, Second-Chance replacement, 7+ kernel system calls, custom fault handler -- reducing page faults by 25%. Built VectorLift, a production-grade semantic search engine implementing dense vector retrieval over FAISS and Qdrant with Kafka-driven indexing pipelines. Designed fault-tolerant, horizontally scalable event-driven systems handling 2,000 async operations/second with zero state loss using Temporal and Kafka. Strong foundation in algorithms, data structures, operating systems, and concurrency.

CORE COMPETENCIES

C / Systems Programming

Operating Systems + Kernel

Vector Index (FAISS + Qdrant)

Concurrency + Multithreading

Distributed Fault-Tolerant Systems

Apache Kafka + Event-Driven

Algorithms + Data Structures

Docker + Kubernetes

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern | Remote

- Designed and shipped **fault-tolerant production systems** using Temporal.io workflow orchestration, multi-model AI pipelines (8+ models), and real-time event streaming -- 394 commits across 7 production deployments, owning features end-to-end.
- Built distributed backend services with TypeScript, Node.js, PostgreSQL, and Redis; implemented role-based access control, document processing pipelines, and S3 file storage for a high-reliability claims automation platform.

NC State University April 2025 - Present

Graduate Research Assistant | Raleigh, NC

- Designed a cloud-based genomic data platform (JBrowse 2 + Google Cloud Run) handling 500GB datasets for 100+ researchers; collaborated on ML architecture with time-series analysis improving model accuracy by 12%.

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern | Remote

- Built AI-driven automation workflows with OCR, NLP, and ML classification models (25% accuracy improvement); automated deployments on AWS and GCP with Grafana monitoring and model reliability testing.

Capgemini Jun 2023 - Aug 2023

AI/ML Intern | Remote

- Built **distributed data pipelines** using Apache Hadoop + Spark for batch processing and **Apache Kafka** for real-time streaming -- reduced ticket-processing latency by 20% and manual processing time by 15%.

PROJECTS

Demand Paging -- Xinu OS Virtual Memory System C / Systems Programming

Engineered an 8-frame virtual memory system in **C** for Xinu OS from scratch: Second-Chance page replacement algorithm, custom page fault handler, inverted page tables, and 7+ new kernel system calls for memory management. Achieved 25% reduction in page faults and 15% memory performance improvement. Work required deep understanding of **operating system internals, memory architecture, and kernel-level concurrency** -- directly extending the Xinu OS codebase used in graduate OS coursework at NC State.

C · Xinu OS · Virtual Memory · Kernel System Calls · Page Tables · OS Internals

VectorLift -- Semantic Search + Ranking Engine Vector Index / LLM

Production-grade search system built on MS MARCO implementing BM25 (Elasticsearch), **dense vector retrieval** over **FAISS and Qdrant** (bi-encoder fine-tuned with PyTorch), and cross-encoder re-ranking. Kafka-driven indexing pipeline for event-based corpus ingestion; FastAPI serving API with per-stage latency tracking; evaluated 6 pipeline configurations with NDCG@10, MRR@10, MAP, and paired bootstrap significance testing. Full stack deployed via Docker Compose with PostgreSQL, Redis, Prometheus, and Grafana.

Python · PyTorch · FAISS · Qdrant · Elasticsearch · Kafka · FastAPI · PostgreSQL · Redis · Docker | github.com/Aditya-k24/VectorLift

AlgoPulse -- Fault-Tolerant Agentic AI Platform Distributed / Fault Tolerance

Event-driven AI orchestration platform scaled to **2,000 async inference requests/second with zero state loss**. Diagnosed API-polling bottleneck under load; redesigned to Supabase Edge Functions streaming with **Temporal.io** for durable workflow state management and **Kafka** for fault-tolerant synchronization across workers. Fault tolerance was a first-class constraint -- any worker crash recovers state from the Temporal event log automatically.

TypeScript · Node.js · Temporal.io · Kafka · Supabase Edge Functions | github.com/Aditya-k24/AlgoPulse

Inventory Reservation Service Concurrency / Server Architecture

Transactional Java 21 + Spring Boot API preventing race conditions via **PostgreSQL row-level locking** (SELECT FOR UPDATE) across concurrent multi-SKU reservations. Verified with Testcontainers integration tests: two concurrent requests against quantity 1 produce exactly one success and one 409 Conflict. Idempotent REST endpoints, Flyway migrations, Spring Security RBAC, multi-stage Docker build, GitHub Actions CI.

Java 21 · Spring Boot · PostgreSQL · Testcontainers · Flyway · Docker · GitHub Actions | github.com/Aditya-k24/InventoryReservationService

EDUCATION

MS Computer Science -- NC State University Aug 2024 - May 2026

Courses: Operating Systems, Artificial Intelligence, Data Science, Algorithms, Object Oriented Development, Software Engineering

BTech Computer Science -- Mumbai University Aug 2020 - May 2024

CGPA: 9.32/10 | Courses: OS, DS & Algorithms, Machine Learning, Data Warehousing, Big Data Infrastructure, Software Engineering

SKILLS

Systems: C, Java, Python, TypeScript, SQL -- OS internals, virtual memory, kernel programming, multithreading, concurrency

Vector/Search: FAISS, Qdrant, Elasticsearch, sentence-transformers, PyTorch, dense retrieval, re-ranking

Distributed: Apache Kafka, Apache Spark, Hadoop, Temporal.io, fault-tolerant event-driven architecture

Backend/Infra: Spring Boot, FastAPI, Node.js, PostgreSQL, Redis, MySQL, MongoDB, Docker, Kubernetes, AWS, GCP

Observability: Prometheus, Grafana, OpenTelemetry, Testcontainers, GitHub Actions CI/CD